Project Overview

Built an end-to-end revenue analysis and financial planning model using an e-commerce behavioral dataset to identify conversion bottlenecks, simulate growth scenarios, and evaluate operational budget implications.

Datset: 3.0M users, 9.2M sessions, 41M views, 743K purchases, $230M

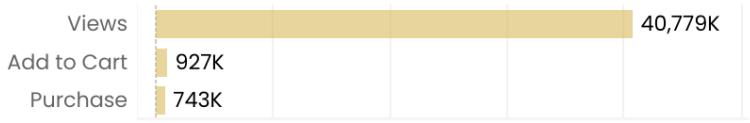
Tools: SQL, Tableau, Excel..

Users	Orders	Sessions	Revenue	AOV
3.0M	**743K**	**9.2M**	**$230M**	**$310**

Key Insights

Revenue performance can be decomposed **Revenue = Sessions × Conversion Rate × AOV**, linking user behavior directly to financial outcomes. Funnel analysis shows the primary bottleneck occurs at the **view → add-to-cart**, where only **2.27% of product views convert to cart additions**, while the **cart → purchase conversion remains strong at 80.18%**. As a result, improving the view-to-cart conversion rate represents the most efficient growth lever because it monetizes existing traffic without requiring proportional increases in acquisition costs. Scenario modeling confirms this dynamic: **conversion improvement drives the largest incremental revenue impact (+$74M)** compared with session growth (+$23M) and AOV improvement (+$12M). Under the growth plan, revenue increases from **$230M to $339M (+$109M)**. Supporting this growth requires operational expansion across **COGS, marketing, logistics, and support**, increasing total costs from **$178M to $273M** and resulting in profit growing from **$52M to $66M (+$14M)**.

Conversion Funnel Analysis



View → Add to Cart Conversion

2.27%

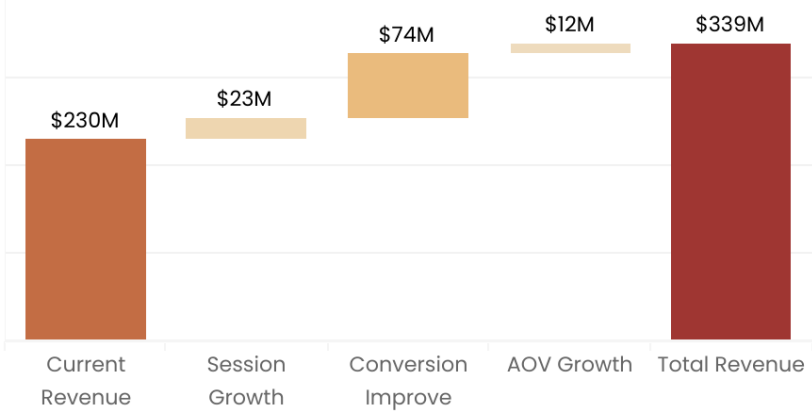
Add to Cart → Purchase Conversion

80.18%

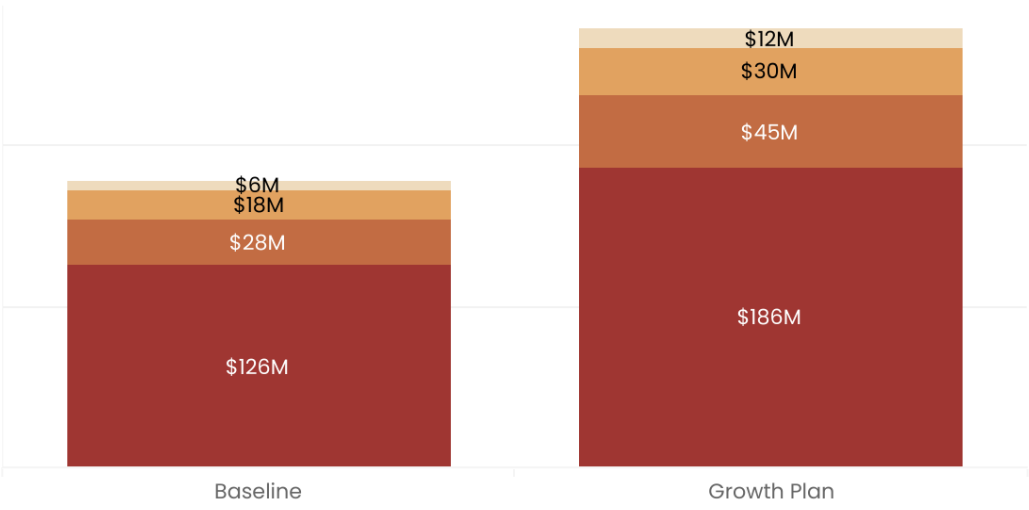
Revenue Growth Scenarios

Scenario	AOV Growth	SessionGro..	ViewCart Lift	Revenue
Baseline				$230M
Sessions + 10%		10%		$253M
Conversion 3%			0.73%	$304M
AOV Growth 5%	5%			$242M
Growth Plan	5%	10%	0.73%	$339M

Revenue Growth Drivers



Operational Budget Impact
Baseline → Growth Plan



Profit

Baseline → Growth Plan

$66M

↑ (+$14M)

$52M